7. Develop an Android application using controls like Button, TextView, EditText for designing a calculator having basic functionality like Addition, Subtraction, multiplication, and Division.

AIM

To design and develop a functional Android application that utilizes UI controls such as `Button`, `TextView`, and `EditText` to perform basic arithmetic operations.

ALGORITHM

1. **UI Design:**
   - Open `activity_main.xml`.
   - Add two EditText fields to accept numerical input from the user.
   - Add four `Button` controls for Addition (+), Subtraction (-), Multiplication (*), and Division (/).
   - Add a `TextView` to display the final calculated result.
2. **Logic Implementation:**
   - In `MainActivity.java`, declare variables for the UI components.
   - Initialize these components using `findViewById()`.
   - Create `OnClickListener` events for each button.
3. **Data Processing:**
   - Retrieve the input strings from the `EditText` fields using `getText().toString()`.
   - Convert the strings to numeric types (e.g., `Double`).
   - Perform the arithmetic operation corresponding to the button clicked.
4. **Output:**
   - Set the calculated value as the text of the `TextView` using `setText()`.
5. **Deployment:** Run the application on an Android Emulator or a physical device to test functionality.

SOURCE CODE

Layout File (activity_main.xml)

XML
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center"
    android:background="#F0F4F8">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Weather Utility: Calculator"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#2C3E50"
```

```
        android:layout_marginBottom="30dp"/>

    <EditText
        android:id="@+id/num1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter first number"
        android:inputType="numberDecimal"
        android:background="@android:drawable/editbox_background"/>

    <EditText
        android:id="@+id/num2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter second number"
        android:inputType="numberDecimal"
        android:layout_marginTop="10dp"
        android:background="@android:drawable/editbox_background"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center"
        android:layout_marginTop="20dp">

        <Button android:id="@+id/btnAdd" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="+" />
        <Button android:id="@+id/btnSub" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="-" />
        <Button android:id="@+id/btnMul" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="*" />
        <Button android:id="@+id/btnDiv" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="/" />
    </LinearLayout>

    <TextView
        android:id="@+id/txtResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result: 0"
        android:textSize="22sp"
        android:layout_marginTop="30dp"
        android:textColor="#E74C3C"/>
</LinearLayout>
```

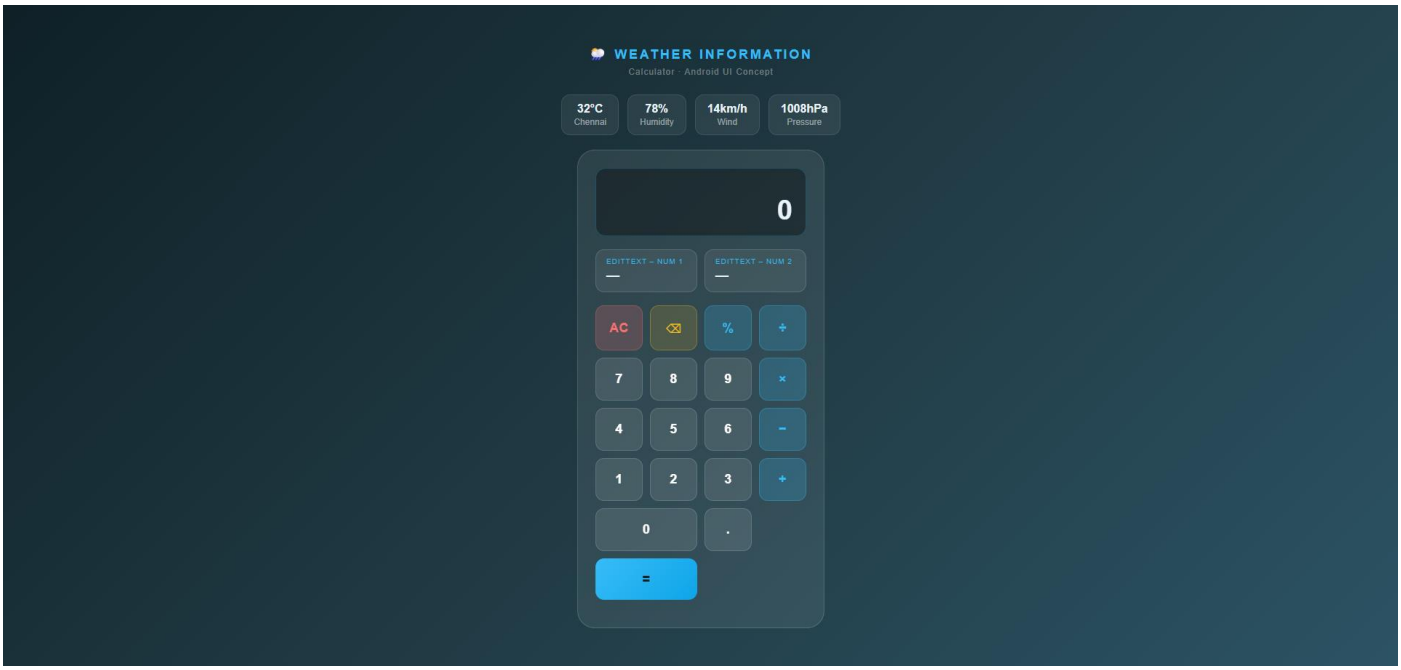
Java File (MainActivity.java)

Java

```
package com.weatherapp.calculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText n1, n2;
    Button add, sub, mul, div;
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        n1 = findViewById(R.id.num1);
        n2 = findViewById(R.id.num2);
        add = findViewById(R.id.btnAdd);
        sub = findViewById(R.id.btnSub);
        mul = findViewById(R.id.btnMul);
        div = findViewById(R.id.btnDiv);
        result = findViewById(R.id.txtResult);

        View.OnClickListener calcListener = new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                double num1 = Double.parseDouble(n1.getText().toString());
                double num2 = Double.parseDouble(n2.getText().toString());
                double res = 0;

                int id = v.getId();
                if (id == R.id.btnAdd) res = num1 + num2;
                else if (id == R.id.btnSub) res = num1 - num2;
                else if (id == R.id.btnMul) res = num1 * num2;
                else if (id == R.id.btnDiv) res = num1 / num2;

                result.setText("Result: " + res);
            }
        };

        add.setOnClickListener(calcListener);
        sub.setOnClickListener(calcListener);
        mul.setOnClickListener(calcListener);
        div.setOnClickListener(calcListener);
    }
}
```

OUTPUT :



RESULT

An Android application was successfully developed using basic UI controls to perform arithmetic operations. The app demonstrated effective event handling and UI-to-logic data binding.